DUNet: A Dorsal Hand Vein Recognition System Using Ensemble Deep Learning Model

Garima Agarwal, Debbrota Paul Chowdhury, Sambit Bakshi

***Abstract*—The challenge in person identification lies in selecting the best biometric trait, as different traits suit different applications. For example, iris recognition is ideal for high-security settings like banking or airport security but is not practical for university attendance. Similarly, fingerprints are commonly used for authentication in security systems, but due to the pandemic, contactless authentication methods have become the new norm. One promising solution is the use of dorsal hand veins for person recognition, as they provide a unique, internal, and contactless method for verifying identity. This technique can be applied to security systems, access control, and financial transactions, where high-level verification is required. This work presents a dorsal hand vein recognition system that uses U-Net as a preprocessing step to enhance accuracy. After segmentation, various deep learning models were applied for recognition, and it was found that ensemble models outperformed individual models. By combining multiple models, ensemble approaches leveraged the strengths of each, leading to better overall performance in accurately recognizing dorsal hand veins. The system showed a significant improvement in recognition accuracy, with a 17% increase when U-Net was used. The experiment utilized two publicly available datasets: the Dorsal Hand Veins Image Database and Dr. Badawi's Hand Vein Dataset.**

***Index Terms*—dorsal hand vein recognition, U-Net, CNN, Biometrics, ResNet.**

I. Introduction

Biometrics is the measurement and statistical analysis of people's unique physical and behavioral characteristics. The technology is primarily used for identification and access control, or for identifying individuals who are under surveillance. Biometrics involves the use of unique biological traits such as fingerprints, facial recognition, iris patterns, voice recognition, and vein patterns for identification purposes. These traits are considered reliable and difficult to forge, making them ideal for security applications. Biometrics is widely used in various fields such as public security, access control, banking, and other critical sectors. Their versatility and reliability make them suitable for a wide range of applications, from securing personal devices to managing large-scale security systems. Biometric technology plays a significant role in automatic personal identification systems, enhancing security and efficiency. The concept of biometrics has been around for centuries, with early forms of biometric identification tracing back to ancient civilizations. However, the modern use of biometrics began in the late 19th and early 20th centuries with the development of fingerprinting techniques. The field continues to evolve with advancements in technology, leading to the sophisticated biometric systems we have today.

Biometric systems offer a higher level of security compared to traditional methods such as passwords or PINs. Since biometric traits like fingerprints, face, palm, ear, and vein patterns are unique to each individual, they are difficult to replicate or forge, making them ideal for secure access control and identity verification. Biometric authentication streamlines the identification process, making it faster and more convenient for users. It also eliminates the need to remember complex passwords or carry identification cards, simplifying access to secure areas or systems. With advancements in technology, biometric systems have become more sophisticated and reliable. Techniques such as machine learning and artificial intelligence are increasingly integrated into biometric systems to enhance their accuracy and adaptability.

Biometric traits can be categorized into two groups: physiological and behavioral. Physiological biometrics are those characteristics that a person is born with and that generally remain stable throughout life like fingerprint, face, iris, ear, dorsal hand vein. Voice, signature, and gait are examples of behavioral biometrics that can be consciously modified by a person.

Each biometric trait has its own advantages and limitations. For example, iris recognition offers high accuracy but involves complex acquisition procedures. Fingerprint patterns may change or fade over time due to aging, injuries, or wear Facial features can be intentionally altered using makeup or accessories, making them less reliable in certain scenarios. To overcome these challenges, this study employs dorsal hand vein images (Figure 1) for personal identification. A comparative overview of commonly used biometric modalities with DHV (Dorsal Hand Vein) biometrics is shown in Table I. Vein recognition is a biometric technique that identifies individuals based on the unique patterns of veins in their bodies. It is widely studied in the research community due to its high security and accuracy in personal identification. Unlike more common biometrics such as fingerprints or facial recognition, vein patterns are located beneath the skin, making them inherently difficult to replicate or alter, which provides a higher level of security. The classical working principle of vein biometrics is illustrated in Figure 2. For acquisition, infrared light is used to scan the veins, which appear as dark lines due to the absorption of light by hemoglobin in the blood. Then the acquired image is preprocessed to make the veins stand out clearly. The unique vein pattern is compared to stored patterns in a database. If a match is found, the individual is successfully

Garima Agarwal, and Debbrota Paul Chowdhury is with the Department of Computer Science and Engineering, National Institute of Technology Silchar, India, and Sambit Bakshi is with the Department of Computer Science and Engineering, National Institute of Technology Rourkela, India, e-mail: dchowdhury@cse.nits.ac.in

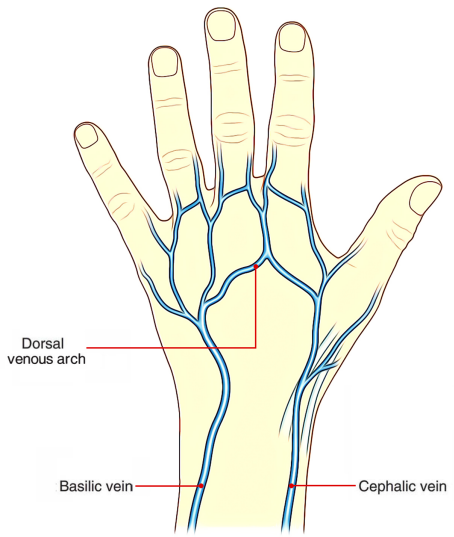

Fig. 1: The dorsal venous network of the hand [1]

recognized otherwise, they are identified as an imposter.

The rest of the paper is arranged as follows: Section II discusses the literature survey and potential research gap in DHV recognition system. The proposed methodology is illustrated in Section III. All the results and findings are discussed in Section IV. Then the conclusion is drawn in Section V.

II. Literature Survey

Vein biometrics begin gaining prominence in the early 2000s. Although biometric systems, which include fingerprints, facial, and iris recognition, have been in development for over a century, vein biometrics are a newer technology that has gained traction in recent decades. Initial research into vein patterns is conducted in the late 1990s, focusing on the use of near-infrared light to capture vein patterns beneath the skin. The technology becomes commercially viable in the early 2000s when companies began to develop and deploy vein pattern recognition systems for various applications, including access control and financial transactions. The high accuracy and security offered by vein biometrics lead to their adoption in high-security environments and continue to evolve with advancements in imaging and machine learning techniques.

Joseph Rice [2] invented hand vein pattern recognition and filed a patent in the United States, marking the inception of DHV biometrics. Later MacGregor et al. [3] introduced *'Vein Check'*, a DHV recognition system aimed at personnel authentication. This system was commercialized through a partnership between Cambridge Consultants and the British Technology Group. The launch represented one of the first practical applications of DHV technology in the market. Further enhancements in DHV technology were made by Cross and Smith [4] using thermographic imaging. The technology began to attract more attention from researchers and developers. Then Alex Hwansoo Choi launched the *BK-100*, the first commercial DHV recognition product [5]. This product marked a significant milestone in making DHV technology accessible to the public. Recent advancements in deep learning have increased the popularity of DHV recognition among researchers. The study [6] develops a dorsal hand vein recognition system using Local Binary Pattern (LBP) features and an Artificial Neural Network (ANN). It processes 240 images with techniques like ROI cropping, mean filtering, CLAHE enhancement, and binarization before feature extraction and classification. The method achieves a 90% recognition rate in training and 100% accuracy in matching results on a test set. In [7] the authors propose a fractal technique for feature extraction from dorsal hand vein images, utilizing the box-counting (BC) method to determine tissue properties. Preprocessing includes ROI extraction, grayscale conversion, CLAHE enhancement, and morphological operations. Experimental results demonstrate the method's effectiveness, yielding promising outcomes compared to existing techniques. Deep Learning (DL) and Generative Adversarial Networks (GANs) are integrated in [8] to improve biometric authentication using dorsal hand veins. The DL-GAN method is tested on two databases achieves high accuracy rates—98.36% and 96.43% surpassing traditional biometric techniques. This approach enhances identity verification by leveraging advanced deep learning and GAN technologies. In [9] the authors propose a novel biometric authentication method by integrating Deep Learning (DL) with Generative Adversarial Networks (GANs) for analyzing dorsal hand vein images. The DL-GAN framework is evaluated on two distinct databases, yielding high accuracy rates of 98.36% and 96.43%, demonstrating superior performance over conventional biometric methods. [10] addresses the lack of a simultaneous dataset for fingerprint and finger vein biometrics by introducing the NUPT-FPV dataset, the first of its kind, collected using a novel integrated device from 140 subjects, providing 33,600 images. A CNN-based multimodal fusion algorithm, FPV-Net, was developed for feature extraction and fusion, leveraging the advantages of simultaneous biometric data collection and interaction to enhance recognition performance. The paper [11] proposes two novel finger-vein recognition models: Rec-FFVN, which uses rectangular filters, and Semi-PFVN, a lightweight network with pretrained layers. Experiments on HKPU, FV-USM, and SDUMLA databases show that rectangular filters enhance accuracy and the Semi-PFVN model, with fewer parameters, outperforms traditional models in recognition accuracy. Experiments on HKPU, FV-USM, and SDUMLA databases confirm this, with Rec-FFVN achieving Rank-1 accuracy of 82.22%, 93.29%, 85.69% and EER of 6.11%, 2.17%, 3.30%, while Semi-PFVN outperforms feature-engineering and DL models with Rank-1 accuracy of 93.05%, 94.67%, 96.61% and EER of 2.47%, 1.37%, 0.99%. In [12] the authors review existing finger-vein recognition methods, focusing on image acquisition, preprocessing, feature extraction, and matching. The article discusses various approaches, including template-based, representation-based, and deep learning methods, and examines the challenges and prospects of finger-vein recognition, particularly in 3D. The paper [13] proposes a PAD method for finger-vein biometrics using DSC with residual connections and LSVM. It applies data augmentation, DSC for feature extraction, and LSVM for classification. Results on Idiap and SCUT-

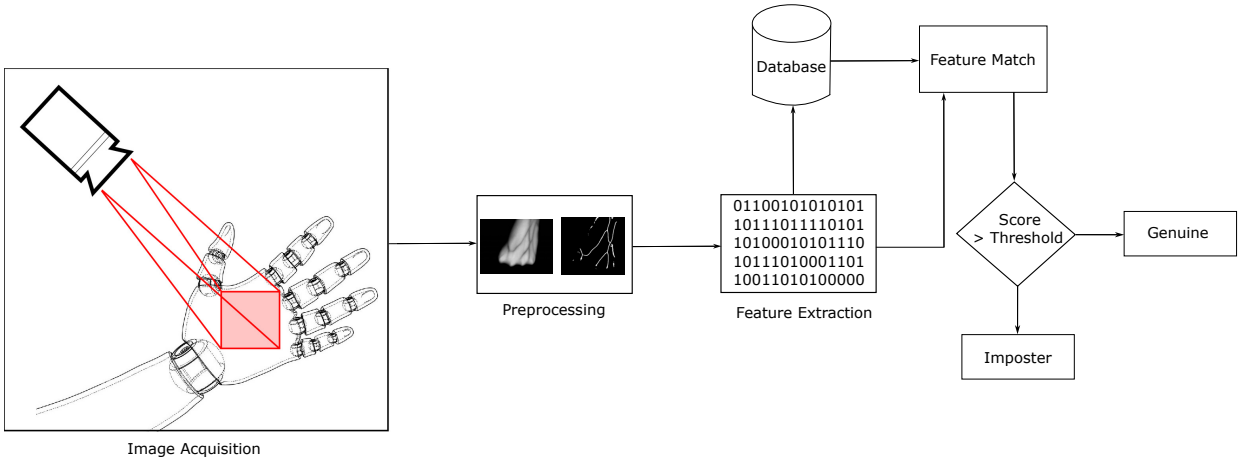

Fig. 2: Generalized Block Diagram of Dorsal Vein Biometric System

FVD datasets show a 0.00% error rate, surpassing existing methods. The paper [14] introduces LE-MSVT, a palm-vein recognition model combining multiscale transformer features and label enhancement via a graph convolutional network. It achieves 98.5% accuracy on the PolyU dataset, outperforming CNNs and existing methods while maintaining efficiency for real-time applications. The paper [15] shows Vision Transformers (ViTs) outperform CNNs in Vascular Biometric Recognition (VBR) using Transfer Learning. ViTs trained on ImageNet-1k/21k achieved high True-Positive Identification Rates: HKPU (99.52%), FV-USM (99.1%), Vera (99.39%), CASIA (96.00%), Bosphorus (99.86%), PUT-wrist (99.67%), and UC3M-CV1+CV2 (99.67%). It also introduces UC3M-CV3, a new contactless wrist vein dataset. The paper [16] introduces FV-ViT, a Vision Transformer model for finger vein recognition, which excels by applying regMLP regularization. FV-ViT achieves state-of-the-art results with EERs of 0.042% and 1.033% on FV-USM and SDUMLA-HMT, respectively, proving its effectiveness even with small datasets. In [17] authors explore a finger vein identification system using PCA for feature extraction and SVM for classification. The study employs three publicly available datasets—THU-FVFDT2, SDUMLA-HMT, and FV-USM—and demonstrates the superiority of the proposed method in terms of recognition rates compared to existing techniques. The methodology includes image acquisition, finger region extraction, pre-processing, feature extraction, and classification. The article [18] presents a dorsal hand vein (DHV) recognition system using image processing and deep learning. It applies CLAHE and Median filtering for preprocessing, followed by CNN-based feature extraction. The CNN model achieved 99.7% accuracy, 97% recall, and 96% precision, while an alternative Restricted Boltzmann Machine (RBM) approach outperformed it with 99.9% accuracy, 99% recall, and 99% precision, with significantly reduced recognition time. The paper [19] presents an LSTM-based Finger Vein Recognition (FVR) model (LSTM-DMT) implemented with TensorFlow and Keras. It achieves 98.93% accuracy, outperforming CNN (95.22%), VGG-16 (95.43%), and ResNet-152 (97.71%). It also has the lowest EER (1.08%) and an efficient 100s computation time, proving its effectiveness for biometric authentication. The study [20] introduces a novel deep learning method for detecting hand wrist creases using custom YOLOv8 models, achieving a high accuracy of 97.1%. This approach aims to improve biometric security systems by leveraging deep learning for more reliable and accurate identification. The paper [21] proposes a fusion approach for palmprint and palm vein recognition using a sparse unsupervised projection algorithm and partial least squares. It enhances feature consistency and integration, achieving superior recognition with EERs of 0.0173%, 0.0192%, 0.0059%, 0.0010%, and 0.0008%. The paper [22] achieves a weighted mean EER of 0.4% and TAR@FAR=0.01 of 99.76% across seven public contactless palm vein databases. It consistently outperforms SOTA models like PalmNet, MPSNet, and Modified U-Net on both EER and TAR metrics. Despite its dual-branch architecture, RSNet maintains competitive computational complexity, with only 11.5% increase in spatial cost compared to AMPVNet.

Recently, the biometric market has seen a rise in products based on hand and vein traits, with Fujitsu's *PalmSecure* system being notable. Despite three decades of research, DHV biometrics have not achieved widespread adoption, with limited success in products. The *VP-II X Hand Vascular Scanner* is recognized for its fast verification speed and high accuracy, demonstrating the potential of DHV technology in practical applications. The literature review reveals that there are very few publicly available datasets for dorsal hand veins. In most of the papers, authors have used Dr. Badawi's [23], DHVI [24], and NCUT [25] for dorsal hand vein recognition.

III. Methodology

A. Preprocessing

For this study we use two publicly available datasets
Dataset I : Dorsal Hand Vein Image databse [24]
Dataset II : Dr. Badawi's Hand Vein dataset [26]

1) Automatic Segmentation of Dataset I (Dorsal Hand Veins Image Database): A series of preprocessing steps is applied to the dorsal hand vein images to enhance image quality and improve the accuracy of vein pattern recognition by generating a reliable segmentation mask. First, the image contrast is

Biometric Traits	Advantages	Disadvantages
Fingerprint	• **Cost-Effectiveness:** Fingerprint devices are affordable with low maintenance, making them ideal for access control and biometric authentication. • **Compact Size:** Fingerprint authentication devices are small and portable, making them easy to integrate into different environments. • **User-Friendliness:** Fingerprint scanning is rapid, simple, and requires little training, making it easy to adopt in a variety of contexts. • **Wide Availability:** Fingerprint devices are inexpensive, widely accessible, and simple to incorporate into current security systems.	• **Easy Replication:** Fingerprints left on surfaces (e.g., glass or door handles) can be lifted, enhanced, and used to create fake fingerprint readers, making them vulnerable to hacking or unauthorized access. • **Impact of Injuries and Wear:** Fingerprints can be affected by injuries, cuts, or worn-out skin, leading to identification errors in biometric systems. • **Environmental Sensitivity:** Environmental factors like dirt, moisture, and temperature changes can degrade fingerprint quality, causing authentication errors. • **Contact-Based Acquisition:**Fingerprint recognition requires physical contact, raising hygiene concerns in shared or public environments.
Iris	• **Non-Invasive:** Iris recognition is contactless, ensuring user comfort and hygiene. • **Stability:** The iris remains unchanged throughout life, providing long-term reliability. • **Difficult to Forge:** Unique and highly randomized iris patterns offer strong security against spoofing.	• **Lighting Sensitivity:** Poor lighting can degrade image quality, affecting iris recognition accuracy. • **Eyewear Interference:** Glasses or sunglasses must be removed, causing inconvenience for users. • **Controlled Environment Requirement:** Requires particular lighting and placement, which restricts use in dynamic environments.
Face	• **Passive Authentication:** It requires minimal effort from the user, enhancing convenience and accessibility. • **Ease of Deployment:** Integration with existing systems is easy, as face images are already part of many databases like passports and driver's licenses. • **Real-Time Performance:** It enables live surveillance and instant authentication. • **Low Maintenance:** It requires less maintenance than security systems that rely on hardware.	• **Lighting and Background Sensitivity:** Poor lighting or cluttered backgrounds can reduce accuracy and may cause misidentification. • **Viewpoint Sensitivity:** Significant changes in the viewing angle (e.g., profile vs. frontal view) can drastically reduce the accuracy of facial recognition systems. • **Changes in Appearance:** Aging, facial hair, makeup, or accessories (e.g., hats, masks) may impact system performance.
DHV	• **Non-Contact Acquisition:** It does not require physical contact, eliminating hygiene concerns. • **Enhanced Security:** The vein patterns in the hand are internal and unique to each individual. making them difficult to replicate or forge. • **High Contrast Imaging:** Near-infrared (NIR) light enhances vein pattern visibility by creating high contrast with surrounding skin. This ensures clear imaging, even with surface imperfections. • **Minimal Surface Impact:** Vein patterns are not affected by surface conditions such as scars, moles, or dirt. • **Low Impact of Environmental Factors:** It is less affected by environmental conditions like lighting and temperature changes. • **Reliable & User-Friendly:** Stable vein patterns ensure long-term accuracy, and DHV acquisition is fast, simple, and hygienic.	• **Higher Cost of Devices:** DHV devices are more expensive than fingerprint or facial recognition, making them less viable for small businesses. • **Limited Availability of Scanners:** DHV technology is rare, with expensive and limited scanners, restricting accessibility and adoption. • **Less Familiarity:** A lack of awareness about DHV technology may cause resistance and a learning curve for adoption. • **Environmental Sensitivity:** Though resistant to environmental factors, DHV recognition can be affected by temperature, lighting, and hand positioning. These factors may impact scan quality and identification accuracy.

TABLE I: Comparison of commonly used biometrics with DHV

enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). This technique applies localized histogram equalization to increase the visibility of veins against the background while minimizing noise amplification in uniform regions. Common CLAHE parameters include a clip limit of 2.0 and a grid size of (8, 8), which control the extent of local contrast enhancement. Following contrast enhancement, the image is segmented into foreground and background regions using binary thresholding. Several threshold values were tested on Dataset I, and a value of 127 produced the most effective separation between the vein patterns and the background. The maximum intensity value is set to 255, resulting in a binary image that highlights the vein structures. The binary image is processed with morphological operations to enhance the segmented output. Erosion, using a 3×3 kernel, removes small noise artifacts, while dilation reconnects fragmented vein structures, improving their continuity. Skeletonization then reduces the vein structures to a one-pixel-wide representation, preserving their topological features. High-frequency noise is reduced using a Gaussian Blur with a kernel size of (5, 5) and a sigma of 0 to smooth the image. Subsequently, Canny Edge Detection is applied to detect the vein boundaries by identifying abrupt variations in intensity. This method highlights the edges of the vein patterns. For optimal edge detection, a lower threshold of 100 captures finer details, while a higher threshold of 200 emphasizes strong vein

boundaries, helping to minimize noise. This preprocessing pipeline enhances the dorsal hand vein images, resulting in a distinct segmentation mask. Figure 3 illustrates the step-by-step preprocessing, refining vein structures for accurate segmentation.

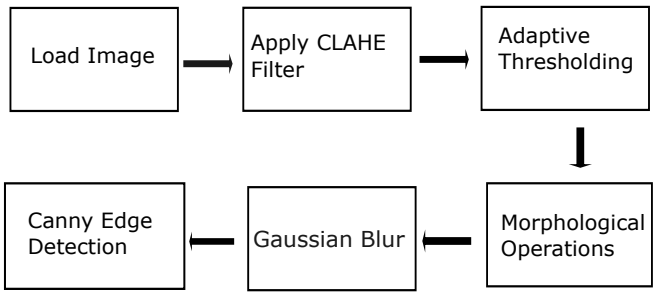

Fig. 3: Preprocessing steps for Dataset I

2) Automatic Segmentation of Dataset II (Dr. Badawi's Hand Vein Dataset): A series of preprocessing steps is applied to Dr. Badawi hand vein dataset to enhance image quality and improve the accuracy of vein pattern recognition by generating a reliable segmentation mask. Initially, all images were resized to 224 × 224 pixels to maintain uniform dimensions across the dataset. To enhance contrast, Contrast Limited Adaptive Histogram Equalization (CLAHE) is employed, which improves the visibility of vein structures by adapting local contrast while minimizing noise amplification. A clip limit of 1.5 and a tile grid size of 8 × 8 are used to balance enhancement and prevent artifacts. For segmentation, adaptive thresholding using the Gaussian method is applied with a block size of 21 and a constant value of C = 2. This technique dynamically adjusts the threshold based on local pixel intensity variations, ensuring a clear separation of veins from the background, even in regions with uneven illumination. Following thresholding, morphological dilation is performed using a 3 × 3 kernel, which helps reconnect broken vein segments and strengthen continuity in vein patterns. This is followed by Canny edge detection, which identifies sharp intensity changes to accurately delineate vein boundaries, using lower and upper thresholds of 50 and 150, respectively. As a final step, skeletonization is applied to reduce the vein structures to a one-pixel-wide representation while preserving their topological features. The structured preprocessing pipeline enhances Dr. Badawi's hand vein dataset for segmentation. Figure 4 illustrates the step-by-step preprocessing, refining vein structures for accurate segmentation.

After obtaining the initial segmentation masks through traditional image processing techniques for both datasets, a deep learning approach is employed to further improve segmentation accuracy and consistency. Traditional methods, although effective in extracting vein patterns, may struggle with complex backgrounds, illumination variations, or low-contrast regions. To address these limitations and capture more intricate vein structures, we integrate a U-Net-based deep learning model into the segmentation pipeline.

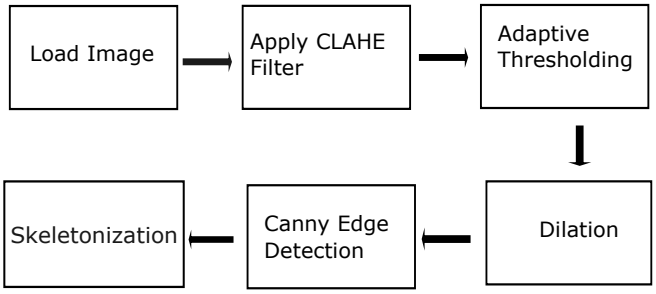

Fig. 4: Preprocessing steps for Dataset II

B. U-Net architecture

The U-Net architecture is first introduced in [27] as a U-shaped convolutional neural network, initially developed for biomedical image segmentation tasks. Its symmetrical encoder-decoder design, along with skip connections (Figure 5), enables the preservation of spatial information throughout the network. These skip connections allow the network to incorporate fine-grained features from earlier layers, thereby enhancing its effectiveness in pixel-wise prediction tasks such as image segmentation.

The input to the U-Net architecture consists of two images: the original dorsal hand vein images and the corresponding preprocessed images. To train the U-Net model for dorsal hand vein segmentation, we split the dataset into 80% for training and 20% for testing to ensure balanced evaluation. The Adam optimizer is used with a learning rate of 0.0001 over 50 epochs. A ResNet–U-Net architecture is employed, integrating the powerful segmentation capabilities of U-Net with a ResNet backbone for enhanced feature extraction. A crucial step in the training pipeline is data augmentation including shifting, scaling, contrast modifications, random rotations, and flipping along both vertical and horizontal axes. To further diversify the training set, Gaussian noise and blurring are also applied. The output of the U-Net model is a binary mask indicating the segmented regions of the dorsal hand vein pattern, which can be further analyzed for biometric authentication purposes.

1) ResNet–UNet: Our experiments for vein segmentation used the ResNet50 architecture, which considers five different feature map sizes, each being half the size of the previous. The input size is $640 \times 480 \times 1$ pixels. The first convolutional layer uses a kernel of 7×7 with depth 64, stride 2, and padding 3, resulting in an output size of 320×240. This is followed by a max-pooling layer with stride 2 and padding 1, reducing the output to 160×120. At this resolution, there are two residual blocks, each containing 2×2 convolutional layers with 3×3 filters and a depth of 64. The output remains 160×120 with a depth of 64. These four-layered residual blocks are repeated three more times, producing an output of 80×60 and a depth of 128 for the second residual block, followed by the third residual block with an output of 40×30 and a depth of 256, and finally the last residual block with an output of 20×15 and a depth of 512. In our ResNet–UNet architecture, the encoder part corresponds to this modified ResNet50. The

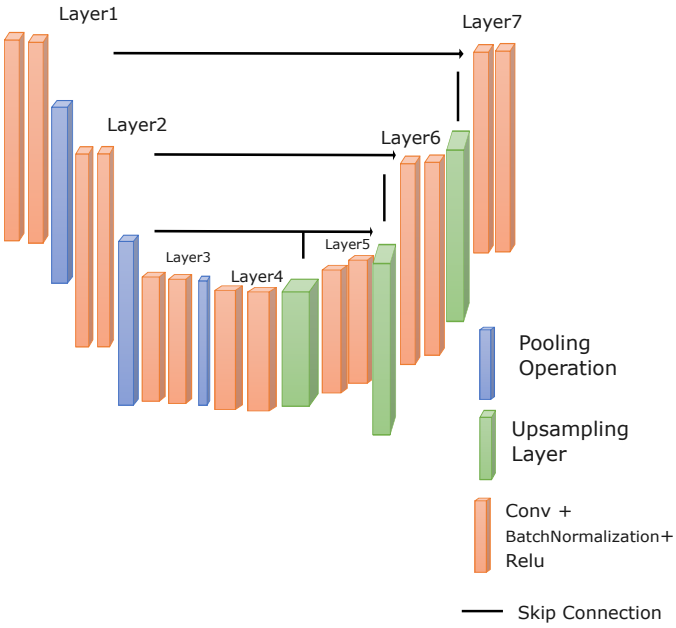

Fig. 5: U-Net architecture

decoder is symmetrical, featuring transpose convolutions and skip connections in a U-Net-like manner.

C. Model Architecture

In this study, multiple deep learning architectures, both individually and in ensemble combinations, are employed to enhance dorsal hand vein recognition. The proposed architecture is depicted in Figure 6. The dorsal image recognition process consists of three stages. The first stage, namely Preprocessing Stage 1, is responsible for the initial vein segmentation. In the second stage, U-Net is implemented for the enhanced vein segmentation. In the third stage, classification models are introduced for recognition purposes. The classification stage uses various CNN models including VGG16, ResNet50, MobileNet, and InceptionV3, each contributing unique strengths. VGG16 extracts fine-grained spatial details through sequential convolutional layers, while ResNet50 leverages residual connections to improve deep feature extraction. MobileNet, designed for efficiency, employs depthwise separable convolutions to reduce computational complexity, making it suitable for real-time applications. InceptionV3, with its multi-scale convolutional filters, captures both local and global vein structures, enhancing feature diversity. To further improve performance, ensemble learning is utilized by combining different CNN models as depicted in Figure 7. Three ensemble architectures are implemented: ResNet + VGG16, which balances deep feature extraction and spatial detail retention; MobileNet + VGG16, which integrates lightweight computation with hierarchical feature learning; and ResNet + MobileNet, which leverages both deep residual learning and computational efficiency.

The proposed deep learning model is trained for 100 epochs with a batch size of 16. A learning rate of $\alpha = 0.0001$ is used to control the model's adaptability during training. The Adaptive Moment Estimation (Adam) optimizer is employed for efficient weight updates. It is an optimization algorithm that dynamically adjusts the learning rate during training to improve convergence. It computes adaptive learning rates for each parameter by utilizing both the first (mean) and second (variance) moments of the gradients, enabling more efficient and stable updates throughout the training process. The parameter update rule in Adam is defined as:

$$w_{t+1} = w_t - \eta \cdot \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \right)$$

with the bias-corrected moment estimates computed as:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

where w_t are the parameters at time step t, η is the learning rate, β_1 and β_2 are the decay rates for the moment estimates (0.9 and 0.999 respectively), ϵ is a small constant (usually 10^{-8}), m_t and v_t are the exponentially decaying averages of past gradients and squared gradients, respectively, $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected versions of m_t and v_t.

The Categorical Cross-Entropy loss function is used to evaluate classification errors. It is calculated using the following formula:

$$\text{Loss} = -\sum_{i=1}^{N} y_i \log(p_i)$$

where N is the number of classes, y_i is the true label, p_i is the predicted probability for class i from the model.

For activation functions, the model utilizes the ReLU (Rectified Linear Unit) activation function in all hidden layers, to introduce non-linearity:

$$\text{ReLU}(x) = \max(0, x) \tag{1}$$

This facilitates the learning of complex vein patterns from the input image data. The final dense layer uses the Softmax activation to output class probabilities for multi-class classification, defined as:

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^{N} \exp(z_j)} \tag{2}$$

where $\hat{y}_i$ is the predicted probability for class i, z_i is the input logit for class i, and N is the total number of classes.

To prevent overfitting and promote training stability, batch normalization is applied after each fully connected layer, and a dropout rate of 0.4 is introduced afterward. The model includes two fully connected (dense) layers. The first dense layer contains 1024 units, and the second contains 512 units. The final output layer consists of $N = 50$ classes for the Dr. Badawi's dataset, and is adjusted to $N = 138$ when applied to the DHVI dataset.

D. Metrics used for the evaluation of the model

The performance evaluation of a classifier is determined using the Precision, Recall, F1-score, and Accuracy. In dorsal hand vein-based identification, the classifier produces four possible outcomes: True Positives (TP) occur when the system

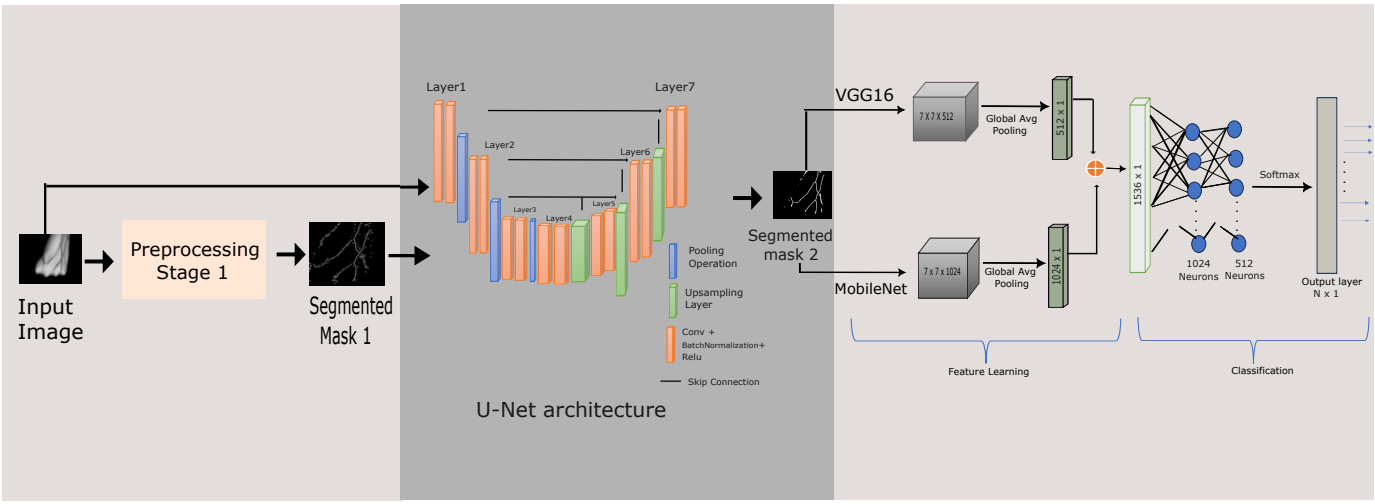

Fig. 6: Proposed Dorsal Hand Vein Recognition System

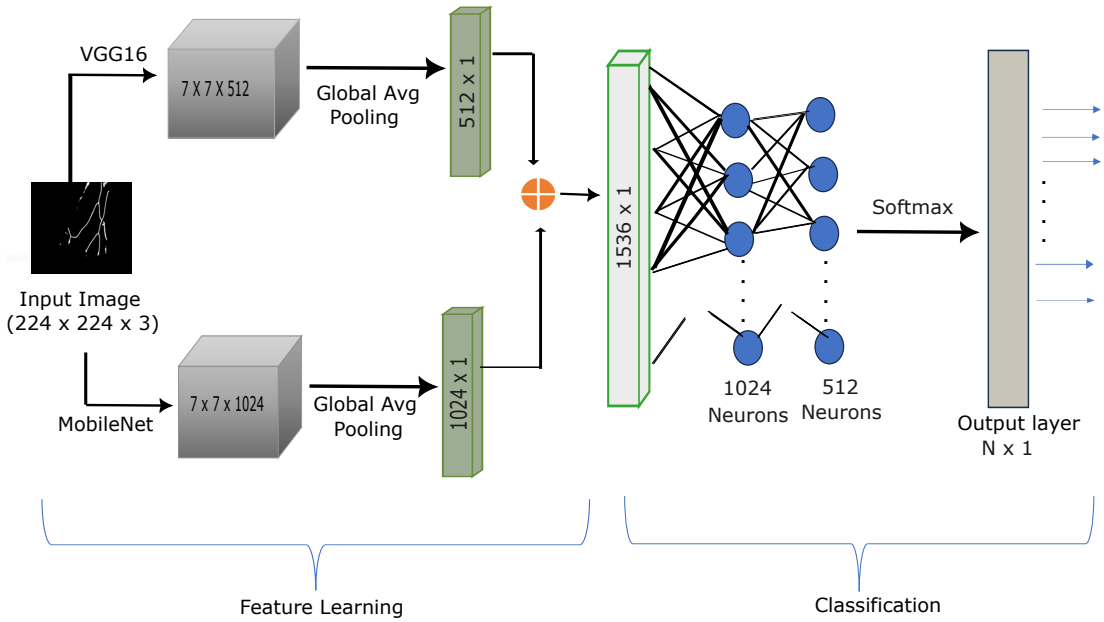

Fig. 7: Proposed Classification Architecture

correctly verifies an individual's identity, while True Negatives (TN) indicate correctly rejecting an unauthorized individual. Conversely, False Positives (FP) occur when the system incorrectly accepts an unauthorized individual as genuine, and False Negatives (FN) occur when an authorized individual is wrongly rejected. These outcomes are used to measure the classifier's accuracy, precision, recall, and overall reliability in vein-based biometric authentication.

- **Accuracy**: Measures how well the model correctly classifies vein patterns by calculating the proportion of correctly identified cases. It provides an overall assessment of the system's reliability.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**: Indicates the proportion of correctly predicted vein patterns out of all the positive predictions made by the model. High precision means fewer false positives.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity)**: Measures the model's ability to correctly detect all actual vein patterns. A high recall reduces the chances of missing valid vein regions.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score**: Represents the harmonic mean of precision and recall. It is useful when there is an imbalance between classes and provides a balanced view of the model's performance.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

IV. Results and Experiments

A. Dataset Used

This study utilizes two distinct datasets for hand vein recognition. The first dataset, referred to as the **Dorsal Hand Veins Image Database** [24], consists of two parts. Database 1 includes 138 individuals, with four images per person per hand, totaling 1,104 images. Each image is named in the format `person_[xxx]_db1_[yz].tif`, where $[xxx]$ is the person's ID (001–138), $[y]$ indicates the hand (L or R), and $[z]$ denotes the image number (1 to 4). Database 2, which includes 113 individuals, has three images per person per hand, resulting in 678 images. This database follows a similar naming convention: `person_[xxx]_db2_[yz].tif`, where $[xxx]$ is the person's ID (001–113), $[y]$ denotes the left (L) or right (R) hand, and $[z]$ is the image number (1 to 3). A two-month gap exists between the data collection sessions for Database 1 and Database 2.

The second dataset, **Dr. Badawi's Hand Vein Dataset** [24], consists of 500 images from 50 individuals, with five images per person per hand. Both the left and right hands of the individuals are included in the dataset, the left hands are represented by IDs 1 through 50, and the right hands are represented by IDs 51 through 100. Since the dataset is intended for identification purposes, differences between the left and right hands can be analyzed.

B. Findings

All the experiments to identify dorsal hand veins are performed using two publicly available datasets: the Dorsal Hand Veins Image Database [24] and Dr. Badawi's Hand Vein Dataset [26]. Accurate segmentation of the veins is essential for building an effective dorsal hand vein recognition system. To enhance segmentation quality, the proposed algorithm follows a two-stage preprocessing approach. In the first stage, the input image undergoes a sequence of operations, including CLAHE, binary thresholding, morphological operations (erosion, dilation, skeletonization), Gaussian blur, and Canny edge detection. The preprocessing pipeline for Dataset-I and Dataset-II is illustrated in Figures 8 and 9, respectively.

As observed in Figures 8 (f) and 9 (f), the segmented vein images are not entirely clear. To improve clarity of the segmented image, an additional preprocessing stage is applied using U-Net to enhance and refine the segmentation. In this stage, both the original image and the output from the first preprocessing stage are used as inputs. The segmented vein image from the first stage serves as the ground truth, while the enhanced segmented vein image generated by U-Net acts as the predicted mask for the original input. The process of this second preprocessing stage is illustrated in Figures 10 and 11 for Dataset-I and Dataset-II respectively.

The predicted masks serve as input to the recognition task. To accurately identify dorsal veins, a classification module is integrated after the preprocessing steps. For classification, various CNN models are implemented, and to improve recognition accuracy, ensemble models are employed as classifiers. To enhance the robustness of the models, the dataset is split

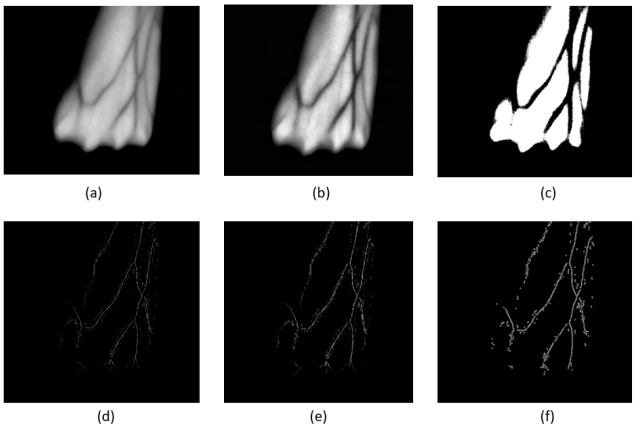

Fig. 8: a) Load Image b) Apply CLAHE filter c) Apply Binary thresholding d) Apply Morphology operations (Erosion, Dilation, Skeletonization) e) Apply Gaussian Blur f) Apply Canny edge detection

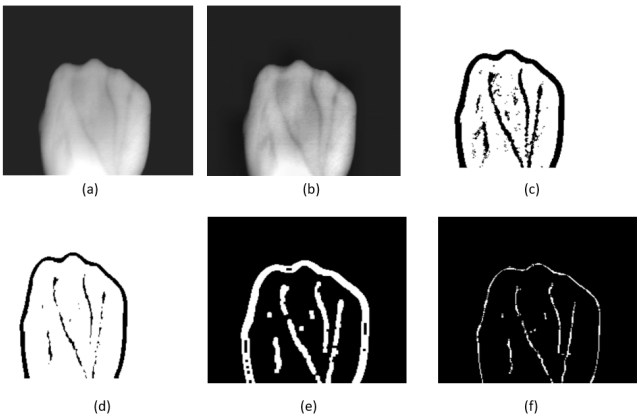

Fig. 9: a) Load Image b) Apply CLAHE filter c) Apply Adaptive thresholding d) Apply Dilation e) Apply Canny edge detection f) Apply Skeletonization

Fig. 10: Preproceesing Stage-II: Enhanced Vein Image Using U-Net on Dataset-I

Fig. 11: Preproceesing Stage-II: Enhanced Vein Image Using U-Net on Dataset-II

Model	Train-Val-Test Split	Precision	Recall	F1-Score
VGG16	80-10-10	74.0	72.6	70.0
	70-10-20	73.8	71.4	70.1
	60-20-20	61.2	58.8	57.2
ResNet50	80-10-10	83.6	82.5	80.4
	70-10-20	81.8	78.8	78.0
	60-20-20	71.6	70.6	67.7
MobileNet	80-10-10	84.1	83.3	81.5
	70-10-20	80.1	79.1	77.2
	60-20-20	74.1	72.1	70.3
InceptionV3	80-10-10	58.6	59.5	55.3
	70-10-20	68.4	64.6	62.0
	60-20-20	53.3	50.5	48.0
Ensemble NN (ResNet+ VGG16)	80-10-10	86.8	85.3	83.6
	70-10-20	81.2	78.0	76.8
	60-20-20	74.2	72.4	70.4
Ensemble NN (MobileNet+ VGG16)	80-10-10	86.9	84.5	83.1
	70-10-20	86.3	84.1	83.6
	60-20-20	69.9	67.8	65.5
Ensemble NN (ResNet+ MobileNet)	80-10-10	88.6	87.3	85.9
	70-10-20	88.4	85.7	84.3
	60-20-20	82.3	79.9	78.6

TABLE II: Analysis of Dataset-I in terms of precision, recall, and F1 score without using UNet

Model	Train-Val-Test Split	Precision	Recall	F1-Score
VGG16	80-10-10	62.2	72.0	65.1
	70-10-20	68.6	71.0	66.5
	60-20-20	62.0	62.0	59.1
ResNet50	80-10-10	69.0	78.0	71.7
	70-10-20	74.5	75.0	72.1
	60-20-20	60.1	61.0	57.0
MobileNet	80-10-10	88.0	92.0	89.3
	70-10-20	69.9	71.0	67.4
	60-20-20	68.3	67.0	64.3
InceptionV3	80-10-10	73.3	80.0	75.3
	70-10-20	70.0	66.0	63.8
	60-20-20	46.3	50.0	44.1
Ensemble NN (ResNet+ VGG16)	80-10-10	85.0	90.0	86.7
	70-10-20	81.0	79.0	77.3
	60-20-20	67.7	69.0	65.4
Ensemble NN (MobileNet+ VGG16)	80-10-10	88.7	92.0	89.7
	70-10-20	77.8	76.0	73.7
	60-20-20	63.2	64.0	60.2
Ensemble NN (ResNet+ MobileNet)	80-10-10	77.5	84.0	79.5
	70-10-20	81.5	80.0	77.7
	60-20-20	73.8	73.0	70.5

TABLE III: Analysis of Dataset-II in terms of Precision, Recall, and F1-score without using UNet

Model	Train-Val-Test Split	Precision	Recall	F1-Score
VGG16	80-10-10	94.8	94.4	93.7
	70-10-20	90.8	89.7	88.8
	60-20-20	90.0	87.9	86.7
ResNet50	80-10-10	97.9	96.8	96.6
	70-10-20	93.4	92.6	91.7
	60-20-20	89.6	87.6	86.7
MobileNet	80-10-10	95.6	95.6	94.9
	70-10-20	95.2	95.5	94.9
	60-20-20	90.8	88.6	87.7
InceptionV3	80-10-10	81.5	80.6	78.5
	70-10-20	81.9	78.3	77.3
	60-20-20	84.2	80.6	79.1
Ensemble NN (ResNet+ VGG16)	80-10-10	96.4	96.8	96.1
	70-10-20	95.1	95.5	94.8
	60-20-20	94.0	92.7	91.6
Ensemble NN (MobileNet+ VGG16)	80-10-10	97.9	98.0	97.6
	70-10-20	96.3	95.8	95.4
	60-20-20	92.3	92.4	91.3
Ensemble NN (ResNet+ MobileNet)	80-10-10	97.0	97.2	96.7
	70-10-20	96.6	96.3	96.0
	60-20-20	94.0	94.2	93.4

TABLE IV: Analysis of Dataset-I in terms of precision, recall, and F1 score with using UNet

Model	Train-Val-Test Split	Precision	Recall	F1-Score
VGG16	80-10-10	88.0	92.0	89.3
	70-10-20	92.8	88.0	87.7
	60-20-20	88.0	83.0	82.8
ResNet50	80-10-10	80.0	86.0	82.0
	70-10-20	94.3	92.0	91.7
	60-20-20	87.8	86.0	84.4
MobileNet	80-10-10	94.0	96.0	94.7
	70-10-20	93.7	91.0	90.5
	60-20-20	92.3	90.0	89.2
InceptionV3	80-10-10	86.0	90.0	87.3
	70-10-20	89.5	84.0	83.8
	60-20-20	76.7	73.0	71.5
Ensemble NN (ResNet+ VGG16)	80-10-10	91.0	94.0	92.0
	70-10-20	96.7	95.0	94.7
	60-20-20	88.3	84.0	82.9
Ensemble NN (MobileNet+ VGG16)	80-10-10	100.0	100.0	100.0
	70-10-20	94.0	90.0	89.6
	60-20-20	87.7	83.0	81.8
Ensemble NN (ResNet+ MobileNet)	80-10-10	91.0	94.0	92.0
	70-10-20	92.0	90.0	89.1
	60-20-20	92.0	88.0	87.0

TABLE V: Analysis of Dataset-II in terms of Precision, Recall, and F1-score using UNet

into various train-val-test ratios, specifically 80-10-10, 70-10-20, and 60-20-20. Performance is evaluated using various metrics, such as precision, recall, F1-score and accuracy. To assess the impact of U-Net on the recognition system, we present two sets of tables for each dataset. One set shows the performance without using U-Net, as presented in Table II and Table III for Dataset-I and Dataset-II, respectively. The other set analyzes the performance with U-Net which is applied to further enhance the segmented dorsal vein image, as shown in Table IV and Table V for Dataset-I and Dataset-II, respectively. By comparing these tables, it is evident that using U-Net significantly improves the performance of the recognition system.

Additionally, Table VI presents the accuracy of the recognition system with respect to various deep learning models used. From the results provided in Tables II, III, IV, V & VI. It is clear that the accuracy of the recognition system is significantly improved when U-Net is incorporated as a preprocessing step. For instance, when using the Ensemble NN model (MobileNet + VGG16), an accuracy of 84.52% is achieved on Dataset-I without U-Net. However, when U-Net is introduced as a preprocessing step, the accuracy increases to 98.61%. Similarly, the accuracy is increased from 92% to 100% for dataset II. After analyzing Table VI, it is observed that the average increase in accuracy after applying the U-Net architecture is approximately 17%. The maximum accuracy improvements are recorded as 30.05% for Dataset-I and 39.99% for Dataset-II, when InceptionV3 is used as the classifier. In contrast, the accuracy difference was minimal when MobileNet was used. Nonetheless, in most cases,

the inclusion of U-Net resulted in a significant increase in recognition accuracy.

For both Dataset I and Dataset II, the model without U-Net shows moderate training and validation accuracy along with higher loss values, indicating limitations in feature extraction. In contrast, integrating U-Net significantly improves accuracy and reduces loss, demonstrating better feature refinement and learning capability. This validates the effectiveness of U-Net in enhancing the ensemble model's performance for dorsal hand vein pattern recognition (see Figure 12 and Figure 13). The proposed methodology has been compared with the existing state-of-the-art in Table VII. From Table VII, it is found that our model outperforms the others work without using any augmentation. Most papers use augmentation to achieve the state-of-the-art accuracy, while our ensemble model with U-Net achieves the same accuracy with the raw dataset.

V. Conclusion

In this work, a dorsal hand vein recognition system is proposed. For the experiments, two publicly available datasets are used: Dataset-I, the Dorsal Hand Veins Image Database, and Dataset-II, Dr. Badawi's Hand Vein Dataset. This study uses a two-stage preprocessing pipeline where U-Net plays a key role in enhancing vein segmentation. The impact of U-Net on model performance is evident, improving feature extraction and overall recognition accuracy. The experimental results demonstrate that the Ensemble NN (MobileNet + VGG16) outperformed individual models, achieving 98.61% accuracy for Dataset I and 100% accuracy for Dataset II. The model also attains high recall and precision values, with Dataset I achieving 97.9% recall, 98.0% precision, and an F1-score of 97.6%, while Dataset II records 100% recall, 100% precision, and an F1-score of 100%.

Notably, the proposed approach achieves state-of-the-art performance without relying on data augmentation, which is commonly used in related work. The integration of U-Net significantly enhances the visibility and segmentation of vein patterns, leading to more reliable feature extraction. Due to its high accuracy and modular architecture, the system shows promise for real-world biometric security applications. In the future, the model can be extended to handle multi-modal biometric fusion, incorporate lightweight transformer-based architectures, and adapt to more diverse acquisition conditions.

Abbreviation

Acknowledgment

The author expresses sincere gratitude to Dr. Dalton Meitei Thounaojam, Dr. Ramanujam E, Dr. Shyamosree Pal from the Department of Computer Science and Engineering, NIT Silchar, for their valuable comments.

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Model	Train-Val-Test Split	Dataset-1		Dataset-2	
		Test Accuracy (in %) with without UNet	**Test Accuracy (in %)** with UNet	**Test Accuracy (in %)** without UNet	**Test Accuracy (in %)** with UNet
VGG16	80-10-10	72.61	94.44	72.00	92.00
	70-10-20	71.42	89.68	70.99	87.99
	60-20-20	58.79	87.87	62.00	82.99
ResNet50	80-10-10	82.53	96.82	77.99	86.00
	70-10-20	78.83	92.59	75.00	92.00
	60-20-20	70.60	87.62	61.00	86.00
MobileNet	80-10-10	83.33	95.63	92.00	91.99
	70-10-20	79.10	95.50	70.99	91.00
	60-20-20	72.11	88.63	67.00	95.99
InceptionV3	80-10-10	59.52	80.55	80.00	95.49
	70-10-20	64.55	78.30	66.00	83.99
	60-20-20	50.50	80.55	50.00	89.99
Ensemble NN (ResNet+VGG16)	80-10-10	85.31	96.82	89.99	93.99
	70-10-20	78.04	95.50	79.00	94.99
	60-20-20	72.36	92.42	68.99	83.99
Ensemble NN (MobileNet+VGG16)	80-10-10	84.52	**98.61**	92.00	**100**
	70-10-20	84.12	95.76	75.99	89.99
	60-20-20	67.83	92.17	63.99	82.99
Ensemble NN (ResNet+MobileNet)	80-10-10	87.30	97.22	83.99	93.99
	70-10-20	85.71	96.29	80.00	89.99
	60-20-20	79.89	93.93	73.00	87.99

TABLE VI: Comparison of accuracy among deep learning models

Author	Method	Dataset	Train-val-test split	Accuracy
Nour at al. (2024) [18]	CNN		Not given	96.4
Nour at al. (2024) [18]	RBM	Dorsal Hand Vein Image [24]	Not given	96.7
Our	**MobileNet+VGG16**		**80-10-10**	**98.61**
Nour at al. (2024) [18]	CNN		Not given	95.8
Nour at al. (2024) [18]	RBM		Not given	95.5
Masoumeh ZEHTAB NAYEBİ at al. (2021) [28]	CNN	Dr. Badawi's hand vein [26]	80-20	99.64
Norah A. Al-johania1 at al. (2019) [29]	CNN		80-20	100
Our	**MobileNet+VGG16**		**80-10-10**	**100**

TABLE VII: Comparison of the proposed method with others

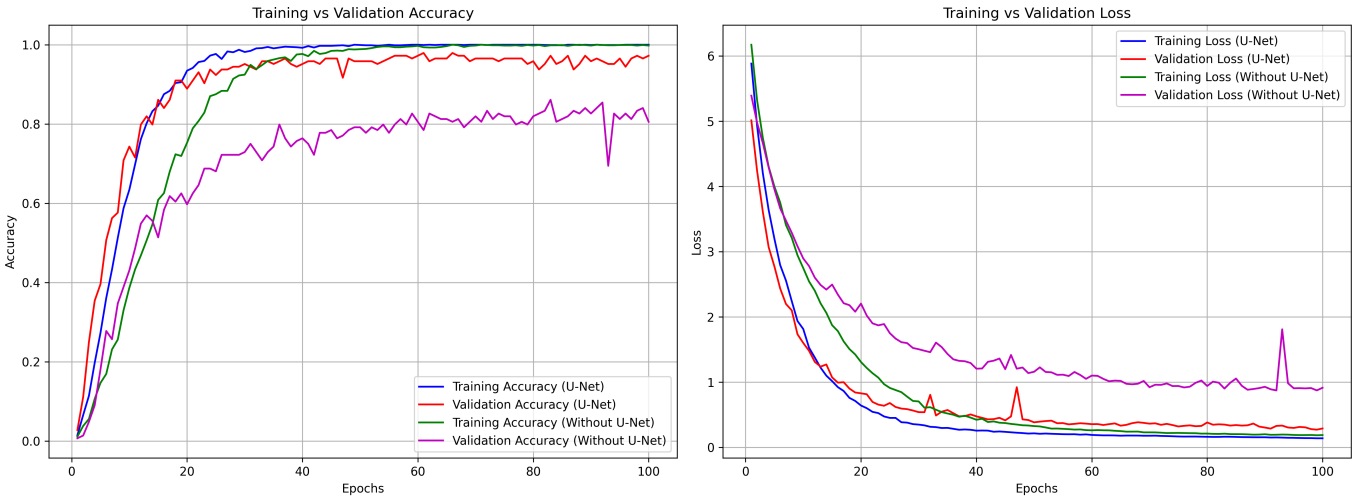

Fig. 12: Performance Comparison of Ensemble NN With and Without U-Net on Dataset I

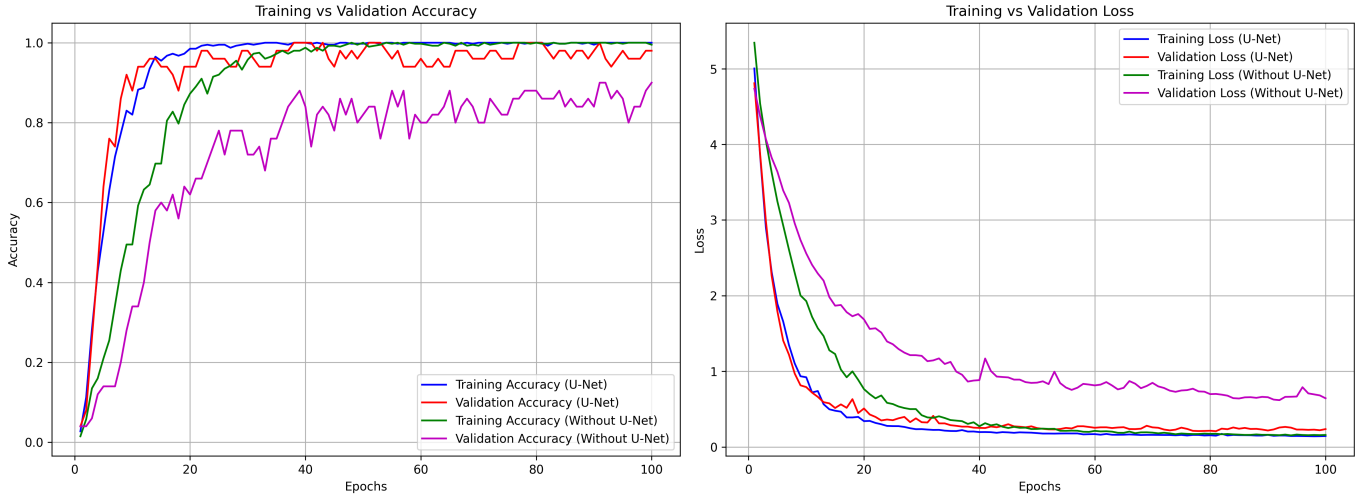

Fig. 13: Performance Comparison of Ensemble NN With and Without U-Net on Dataset II

CNN	:	Convolutional Neural Network
DHV	:	Dorsal Hand Vein
U-Net	:	U-shaped Convolutional Neural Network
RBM	:	Restricted Boltzmann Machine
Ensemble NN	:	Ensemble Neural Network
CLAHE	:	Contrast Limited Adaptive Histogram Equalization
FVR	:	Finger Vein Recognition
LSTM	:	Long Short-Term Memory
DMT	:	Decision-Making Tool
EER	:	Equal Error Rate
CT	:	Computation Time
SVM	:	Support Vector Machine
DL	:	Deep Learning
ViT	:	Vision Transformer
MLP	:	Multi-Layer Perceptron
GAN	:	Generative Adversarial Network
IoU	:	Intersection over Union

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